Flask Backend Developer Handbook

Architecture, Jinja2, Blueprints, Form Handling, & Database CRUD Operations

Target System: SDSF Exam Cell Automation System V2.0
Author & Technical Reference: Joyobrata Das • LinkedIn • GitHub • Portfolio
Designation: M.Tech Artificial Intelligence and Data Science Student • Software Developer (License Owner)
Institution: School of Data Science and Forecasting, Devi Ahilya Vishwavidyalaya, Indore
Contact: +91 8910366038 • **Email:** joyobratadas.85912@gmail.com
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Module 1: The Flask Core Architecture & Request Lifecycle

Flask is a WSGI-compliant microframework built in Python. At its core, Flask maps incoming HTTP requests to Python functions (view functions) and returns HTTP responses.

1.1 The WSGI Request-Response Pipeline

When a client (browser) interacts with Flask, the following sequential pipeline executes:

1. **Request Arrival:** The web server (Werkzeug in development) parses raw TCP/HTTP bytes into a Python `request` object.
2. **Route Matching:** Flask consults its internal URL routing table (the `url_map`) to find a matching endpoint rule.
3. **Pre-Request Processing:** Any registered `before_request` hooks execute (e.g., checking session tokens or opening DB connections).
4. **View Execution:** The mapped view function executes with parameters extracted from the URL.
5. **Response Generation:** The view returns a string, a template render, or a JSON object, which Flask packs into a formal `Response` object (setting status codes and headers).
6. **Post-Request Teardown:** `teardown_request` hooks run (e.g., closing database connections).

1.2 Avoiding Root Package Collisions (`app.py` vs `app/`)

> **Critical Backend Lesson:**
> Never create a root-level file named `app.py` alongside a directory named `app/`. In Python, module resolution will confuse the `app.py` file with the `app/` directory, causing `ModuleNotFoundError: No module named 'app.routes'; 'app' is not a package`.

The standard, clean directory structure for this project is:

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```
SDSF-Exam-Cell-V2.0/
├── app.py                # Main application entry point & WSGI runner
├── routes/               # Blueprint modules (admin, panels, auth)
│   ├── __init__.py
│   └── admin.py
├── templates/            # HTML templates (Jinja2)
│   ├── master_frame.html # Master container layout
│   ├── footer.html       # Shared footer partial
│   └── admin_login_page.html
├── databases/            # Database management
│   ├── connections.py   # Connection factory (TiDB Cloud SSL)
│   ├── schema.sql        # Table DDL definitions
│   ├── init_db.py        # Table initialization runner
│   └── queries.py        # SQL query functions (DML)
└── .env                  # Environment credentials (Git-ignored)
```

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Module 2: Jinja2 Templating Engine & The Master Frame Pattern

Jinja2 is Flask's default template engine. It compiles HTML templates into optimized Python bytecode. Understanding the difference between **Inheritance** and **Inclusion** is vital for building clean UIs without repeating code.

2.1 The Master Frame "Sandwich" Architecture (`master_frame.html`)

Instead of having every child template include the header and footer individually, we use the **Master Frame Pattern**. Think of `master_frame.html` as two slices of bread, and child pages as the filling:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>SDSF Exam Cell Portal</title>
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
</head>
<body class="bg-slate-50 min-h-screen flex flex-col">

    <!-- Top Bread: University Header Banner -->
    <header class="bg-white border-b border-slate-200 shadow-xs">
        <!-- Swapped Logos: SDSF Left, DAVV Right -->
    </header>

    <!-- Filling: Injected Content Area -->
    <main class="flex-1">
        {% block body %}
        <!-- Child template contents are dynamically injected here -->
        {% endblock %}
    </main>

    <!-- Bottom Bread: Automated Footer -->
    {% include 'footer.html' %}

</body>
</html>
```

2.2 Child Template Extension & The Block Scope Rule

Child templates only declare `{% extends 'master_frame.html' %}` and wrap their unique content in `{% block body %}` :

```
{% extends 'master_frame.html' %}

{% block body %}
<div class="flex-1 flex items-center justify-center py-12 px-4">
    <form method="POST" action="/admin/login">
        <!-- Only the login form inputs live here -->
    </form>
</div>
{% endblock %}
```

> **Critical Jinja2 Rule:**

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> When a template uses `{% extends %}`, **never** write HTML outside of a `{% block %}` tag! Any HTML or `{% include %}` placed outside a block is evaluated before the master template and gets pushed to the very top of the generated document above the `<!DOCTYPE html>` declaration.

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2.3 Complete Guide: `render_template()` vs. `redirect()` (Data Transmission & Best Combinations)

In Flask, two core functions control what happens at the end of a view function: `render_template()` and `redirect()`. Beginners often confuse them, but they operate at fundamentally different layers of the HTTP protocol:

Feature / Dimension	`render_template('page.html', ...)`	`redirect(url_for('...'))`
HTTP Status Code	`200 OK` (Normal successful HTML delivery)	`302 Found` / `303 See Other` (Temporary redirect)
Browser Address Bar	**Unchanged.** Stays on the currently requested URL.	**Changes.** Updates immediately to the new target URL.
How It Works Under the Hood	The server opens the `.html` file, executes Jinja2 logic, injects Python variables, and sends raw rendered HTML text back in the HTTP response body.	The server sends an empty body with a `Location: /new-url` response header. The client browser reads this header and automatically issues a brand new `GET` request.
Variable Scope & Memory	Direct access to Python variables passed as keyword arguments.	Destroys current Python local variables. A new request cycle begins from scratch.
Browser Refresh (F5) Behavior	If returned after a POST request, pressing F5 re-submits the form data (causing duplicate database inserts!).	Pressing F5 cleanly reloads the destination GET route safely without duplicating any form action.
Primary Purpose	Displaying an HTML view with freshly queried or calculated data.	Navigating the user to a new location after completing an action (login, insert, update, delete).

Part 1: Passing Values Using `render_template()`

Because `render_template()` executes Jinja2 directly on the server, you can pass rich Python objects (strings, numbers, booleans, dictionaries, lists of dictionaries, and even helper functions) into the template as keyword arguments:

1. Direct Keyword Arguments

```
@admin_bp.route('/dashboard')
def dashboard():
    admin_name = "Joyobrata Das"
    pending_count = 5
    is_superadmin = True

    # Pass individual variables as key=value arguments
    return render_template('admin_dashboard.html',
                           name=admin_name,
                           pending=pending_count,
                           is_admin=is_superadmin)
```

In Jinja2 (`admin_dashboard.html`):

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```
<h1>Welcome, {{ name }}!</h1>
<p>Pending Panels to Review: <span class="badge">{{ pending }}</span></p>
{% if is_admin %}
    <button class="btn btn-danger">Superadmin Settings</button>
{% endif %}
```

2. Dictionary Unpacking (`**context_dict`)

If you have an existing dictionary (e.g. from a database query using `DictCursor`), you can unpack it directly into the template context:

```
admin_profile = {
    'username': 'joyo_admin',
    'department': 'SDSF',
    'email': 'joyobrata@dauniv.ac.in',
    'role': 'Exam Cell Coordinator'
}

# Unpacking unpacks all keys as top-level Jinja2 variables:
return render_template('profile.html', **admin_profile)
```

In Jinja2: `{{ username }}`, `{{ department }}`, and `{{ email }}` are available directly!

3. Passing Lists of Dictionaries (Database `cursor.fetchall()` Results)

This is the standard pattern for rendering tables and dropdowns:

```
with conn.cursor() as cursor:
    sql = "SELECT id, subject_code, subject_name FROM subjects WHERE semester = %s;"
    cursor.execute(sql, (2,))
    all_subjects = cursor.fetchall()  # List of dicts: [{'id': 1, 'subject_code': 'DS5A-101', ...}]

return render_template('subject_list.html', subjects=all_subjects)
```

In Jinja2:

```
<table class="table">
    <tr><th>#</th><th>Code</th><th>Subject Name</th></tr>
    {% for sub in subjects %}
    <tr>
        <td>{{ loop.index }}</td>
        <td>{{ sub.subject_code }}</td>
        <td>{{ sub.subject_name }}</td>
    </tr>
    {% else %}
    <tr><td colspan="3">No subjects found for this semester.</td></tr>
    {% endfor %}
</table>
```

4. Built-in Automatic Context Variables in Jinja2

Flask automatically injects the following global objects into every template rendered via `render_template()` without needing to pass them manually:

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- `request` : Current HTTP request object (e.g., `request.path` , `request.args.get('search')` , `request.method`).
- `session` : The active user session dictionary (e.g., `{{ session.get('user_id') }}`).
- `g` : Flask application context global object for request-scoped data.
- `url_for()` : The reverse URL lookup function.

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Part 2: Passing Values Across a `redirect()`

> **The Redirect Scope Dilemma:**
> Because `redirect()` sends a `302 Found` header telling the browser to make a brand-new `GET` request, all local Python variables in the current function are immediately lost. You cannot pass Python dictionaries, lists, or template names into `redirect()`!

To pass data across a redirect, you must use one of the four following proven backend techniques:

Technique 1: URL Query String Parameters (via `url_for`)

Ideal for non-sensitive filters, search queries, pagination, or status flags:

```
# Inside Route 1 (Form Submission / Action):
@panel_bp.route('/filter', methods=['POST'])
def apply_filter():
    selected_sem = request.form.get('semester')
    selected_program = request.form.get('program')
    # Append parameters directly into url_for kwargs:
    return redirect(url_for('panel.list_panels', program=selected_program, sem=selected_sem))
    # Browser is redirected to: /panel/list?program=MTech&sem=2

# Inside Route 2 (Destination GET Route):
@panel_bp.route('/list')
def list_panels():
    # Read query parameters from request.args:
    program = request.args.get('program', 'All')
    sem = request.args.get('sem', 1, type=int)

    # Query database using these parameters...
    return render_template('panel_list.html', program=program, sem=sem)
```

Technique 2: Dynamic URL Path Parameters

Ideal for entity identifiers (IDs) following an insertion or selection:

```
# Route 1: Creates new record, fetches new ID
@panel_bp.route('/create', methods=['POST'])
def create_panel():
    new_panel_id = db_insert_panel(...)
    # Pass ID into dynamic route:
    return redirect(url_for('panel.view_panel', panel_id=new_panel_id))
    # Browser is redirected to: /panel/view/45

# Route 2: Destination receives panel_id as a typed URL variable
@panel_bp.route('/view/<int:panel_id>')
def view_panel(panel_id):
    panel = db_fetch_panel_by_id(panel_id)
    return render_template('panel_details.html', panel=panel)
```

Technique 3: One-Time Flash Messages (`flash()` + `get_flashed_messages()`) — The Production Gold Standard

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The standard, idiomatic Flask mechanism to display temporary notifications (success alerts, warnings, errors) after a redirect. Flask encrypts the message into the signed session cookie and automatically purges it after it is rendered once:

```
from flask import flash, redirect, url_for

@admin_bp.route('/login', methods=['POST'])
def admin_login_post():
    username = request.form.get('username')
    password = request.form.get('password')

    admin = verify_admin(username, password)
    if admin:
        session['admin_id'] = admin['id']
        # 1. Flash a success message before redirecting
        flash("Welcome back, Administrator! Login successful.", "success")
        return redirect(url_for('admin.dashboard'))
    else:
        # Flash an error message
        flash("Invalid username or password. Please try again.", "danger")
        return redirect(url_for('admin.admin_login'))
```

In your `master_frame.html` (so it displays automatically on any page):

```
<!-- Flash Message Notification Bar in master_frame.html -->
{% with messages = get_flashed_messages(with_categories=true) %}
    {% if messages %}
        <div class="max-w-4xl mx-auto my-4 px-4">
            {% for category, message in messages %}
                <div class="p-4 rounded-md mb-2 text-sm font-medium
                    {% if category == 'success' %}bg-green-50 text-green-800 border border-green-200
                    {% elif category == 'danger' %}bg-red-50 text-red-800 border border-red-200
                    {% else %}bg-blue-50 text-blue-800 border border-blue-200{% endif %}">
                    {{ message }}
                </div>
            {% endfor %}
        </div>
    {% endif %}
{% endwith %}
```

Technique 4: Client Session Storage (`session['key']`)

Used when complex state must persist across multiple redirects (e.g. multi-step wizard form data, authenticated user IDs):

```
@panel_bp.route('/step1', methods=['POST'])
def panel_step1():
    # Save step 1 selections in session
    session['panel_draft'] = {
        'program_id': request.form.get('program_id'),
        'semester': request.form.get('semester')
    }
    return redirect(url_for('panel.panel_step2'))
```


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Part 3: Production Best Combinations & Decision Matrix

When should you use `render_template()` versus `redirect()`? Follow this architectural decision tree:

Combination 1: Validation Failure → `render_template()` (Preserving User Input)

> **Golden Rule: Never Redirect on Form Validation Failure!**
> If the user entered 15 fields and made a mistake on 1 field, redirecting will wipe out all 15 fields, infuriating the user. Instead, re-render the template directly, passing back the validation error and the submitted `request.form` so inputs retain their typed values!

```
@panel_bp.route('/submit', methods=['GET', 'POST'])
def submit_panel():
    if request.method == 'POST':
        title = request.form.get('exam_title')
        committee = request.form.getlist('committee_members[]')

        # Validation: Committee must have at least 3 members
        if len(committee) < 3:
            # DO NOT REDIRECT! Render template directly passing error and submitted data:
            return render_template('panel_form.html',
                                   error="A minimum of 3 committee members is required.",
                                   form_data=request.form)

        # Validation succeeded: Save to database
        db_save_panel(title, committee)
        flash("Exam panel submitted successfully!", "success")
        # PRG Pattern: Redirect to prevent duplicate submission
        return redirect(url_for('admin.dashboard'))

    # Normal GET request:
    return render_template('panel_form.html', form_data={})
```

In HTML: Pre-populate fields using `form_data`:

```
<input type="text" name="exam_title" value="{{ form_data.get('exam_title', '') }}" required>
```

Combination 2: Successful Mutation → `redirect()` + `flash()` (Post-Redirect-Get)

Whenever a database record is created, updated, or deleted, **ALWAYS** finish with a `redirect()` coupled with a `flash()` message. This prevents the dreaded "Confirm Form Resubmission" dialog when the user presses F5 or browser Back.

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2.4 Jinja2 Syntax Master Reference: Control Blocks (`{% if %}` , `{% for %}`), Filters & The Single-Equals Trap

Jinja2 is a fast, expressive templating engine. Understanding its core syntax blocks, statement types, and safety rules prevents subtle bugs and runtime crashes.

1. The Four Jinja2 Delimiters

Delimiter	Name	Purpose & Description	Example
`{{ ... }}`	Expressions	Evaluates Python expressions and prints the result directly into the rendered HTML output.	`{{ user.full_name }}` `{{ total_marks + 10 }}`
`{% ... %}`	Statements / Control Blocks	Executes control logic such as conditionals, loops, template inheritance, and variable assignments.	`{% if is_admin %}` `{% for sub in subjects %}`
`{# ... #}`	Server-Side Comments	Internal developer notes. **Completely stripped on the server** and never sent to the browser HTML!	`{# Confidential: Only admins see this #}`
`<!-- ... -->`	HTML Client-Side Comments	Standard HTML comment. Sent to browser and visible to anyone inspecting page source.	`<!-- Public layout container -->`

2. The `{% if %}` Conditional Block & Logical Expressions

The `{% if %}` block conditionally renders HTML markup based on Boolean logic:

```
{% if role == 'Superadmin' %}
    <span class="badge badge-purple">Master Access</span>
{% elif role == 'Admin' %}
    <span class="badge badge-blue">Exam Cell Coordinator</span>
{% else %}
    <span class="badge badge-gray">Standard Faculty</span>
{% endif %}
```

THE FATAL SYNTAX TRAP: Single Equals (`=`) vs. Double Equals (`==`)

One of the most frequent beginner bugs in Jinja2 is using a single `=` inside an `{% if %}` tag:

```
<!-- CRITICAL SYNTAX ERROR IN JINJA2 -->
{% if msg = 'login error' %}
    <div class="error">Failed</div>
{% endif %}
```

Why this crashes: A single `=` is an assignment operator in Python/Jinja2. Template control blocks do not permit inline assignments. Flask will fail to compile the template, crashing with:

```
jinja2.exceptions.TemplateSyntaxError: expected token 'end of statement block', got '='
```

The Fix: Always use `==` for comparison, or simply test for truthiness:

```
{% if msg == 'login error' %}   <!-- Proper Comparison -->
{% if msg %}                   <!-- Truthy Check: True if msg is non-empty -->
```


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Truthy and Falsy Evaluations in Jinja2

Like Python, Jinja2 evaluates values for Boolean truthiness without requiring explicit comparison:

- **Falsy Values (Skip the block):** `None`, empty string `""`, empty list `[]`, empty dict `{}`, integer `0`, and `False`.
- **Truthy Values (Execute the block):** Non-empty strings (`"error"`), lists with at least 1 element, non-zero integers, and `True`.

```
<!-- Clean, idiomatic error display in login templates: -->
{% if error %}
    <div class="bg-red-50 border border-red-200 text-red-700 px-4 py-2 rounded mb-4">
        {{ error }}
    </div>
{% endif %}
```

Logical Operators: `and`, `or`, `not`, and `in`

```
{# Combining conditions: #}
{% if is_authenticated and not is_suspended %}
    <a href="/panel/create">Generate New Exam Panel</a>
{% endif %}

{# Membership checks: #}
{% if 'admin_id' in session %}
    <a href="/admin/logout">Logout</a>
{% endif %}

{% if exam_status in ['Approved', 'Submitted'] %}
    <span class="text-green-600">Panel Locked for Edits</span>
{% endif %}
```

3. The `{% for %}` Loop & The Special `loop` Variable

Jinja2 provides powerful iteration capabilities over Python lists, tuples, and database dictionaries:

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```
<table class="w-full border">
    <thead>
        <tr><th>#</th><th>Code</th><th>Subject Name</th><th>Action</th></tr>
    </thead>
    <tbody>
        {% for sub in subjects %}
        <tr class="{{ loop.cycle('bg-white', 'bg-slate-50') }}">
            <td>{{ loop.index }}</td>
            <td>{{ sub.subject_code }}</td>
            <td>{{ sub.subject_name }}</td>
            <td>
                {% if loop.first %}<span class="badge">Primary</span>{% endif %}
            </td>
        </tr>
        {% else %}
        <!-- The else block runs ONLY if the list is completely empty! -->
        <tr>
            <td colspan="4" class="text-center text-slate-400 py-4">No subjects configured.</td>
        </tr>
        {% endfor %}
    </tbody>
</table>
```

The Built-in `loop` Context Object

Variable	Description
`loop.index`	Current loop iteration, **1-based** (1, 2, 3...) — ideal for table serial numbers!
`loop.index0`	Current loop iteration, **0-based** (0, 1, 2...).
`loop.first`	`True` if this is the very first item in the iteration.
`loop.last`	`True` if this is the very last item in the iteration.
`loop.length`	The total number of items in the iterable sequence.
`loop.cycle('a', 'b')`	Alternates between given values every iteration — perfect for zebra striping rows!

4. Essential Jinja2 Filters (`| filter`)

Filters transform expressions on the fly using the pipe (`|`) character:

- `{{ user.name | default('Guest User') }}` : Provides a fallback if the variable is undefined or empty.
- `{{ subjects | length }}` : Counts items in a list or characters in a string.
- `{{ title | upper }}` / `{{ title | lower }}` / `{{ title | title }}` : String capitalization.
- `{{ price | round(2) }}` : Mathematical rounding.
- `{{ html_snippet | safe }}` : Tells Jinja2 **NOT** to HTML-escape this variable (renders raw HTML tags).
  Security Note: Never use `| safe` on user-submitted form data, or malicious actors can execute Cross-Site Scripting (XSS) attacks!

5. Local Variable Assignments: `{% set %}`

You can create or calculate temporary local variables directly inside the template scope:

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```
{% set total_papers = semester_count * 5 %}
<p>Estimated examination papers to print: {{ total_papers }}</p>
```

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2.5 Advanced Multi-Template Architecture: Chained Inheritance, Component Composition (`include`) & Macros

A common question from developers is: "*How do I extend multiple layouts or render more than one reusable piece in a single page?*"

> **The Single Inheritance Rule in Jinja2:**
> In Jinja2, **a template CANNOT extend multiple base templates simultaneously**. You cannot write `{% extends 'frame1.html', 'frame2.html' %}` or place two `{% extends %}` tags in the same file. Jinja2 strictly enforces Single Inheritance.
>
> However, there are **four proven, production-grade architectural patterns** to render and compose multiple components, layouts, and reusable sections together cleanly.

Pattern 1: Multi-Level (Chained) Inheritance (Grandparent → Parent → Child)

This is the standard architectural pattern for portals with multiple sub-systems (e.g. Admin Portal, Faculty Portal, Student Portal):

1. **Level 1 (Grandparent — `master_frame.html`):** Defines the root HTML document, university logos, global Tailwind CSS, automated footer, and an open container block:

```
<!-- master_frame.html -->
<!DOCTYPE html>
<html lang="en">
<head>...</head>
<body class="bg-slate-50 min-h-screen flex flex-col">
    <header>...</header>
    <main class="flex-1">
        {% block body %}{% endblock %}
    </main>
    {% include 'footer.html' %}
</body>
</html>
```

2. **Level 2 (Parent — `admin_base.html`):** Extends `master_frame.html` . It fills `{% block body %}` with a dedicated 2-column Admin layout (Sidebar navigation on the left, Content area on the right), and defines a **new nested block** called `{% block admin_content %}` :

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```
<!-- admin_base.html -->
{% extends 'master_frame.html' %}

{% block body %}
<div class="max-w-7xl mx-auto px-4 py-6 flex gap-6">
    <!-- Persistent Admin Sidebar -->
    <aside class="w-64 bg-white border border-slate-200 rounded-lg p-4 shadow-sm">
        <nav class="space-y-2">
            <a href="/admin/dashboard" class="block py-2 px-3 rounded hover:bg-slate-100">Dashboa
            <a href="/panel/create" class="block py-2 px-3 rounded hover:bg-slate-100">Exam Panel
            <a href="/admin/logout" class="block py-2 px-3 rounded text-red-600 hover:bg-red-50">
        </nav>
    </aside>

    <!-- Nested Dynamic Content Socket -->
    <section class="flex-1 bg-white border border-slate-200 rounded-lg p-6 shadow-sm">
        {% block admin_content %}
        <!-- Child page content will inject here! -->
        {% endblock %}
    </section>
</div>
{% endblock %}
```

3. **Level 3 (Child — `admin_dashboard.html`):** Extends `admin_base.html` (NOT `master_frame.html` directly!). It simply fills the nested block:

```
<!-- admin_dashboard.html -->
{% extends 'admin_base.html' %}

{% block admin_content %}
<h1 class="text-2xl font-bold text-slate-900 mb-4">Admin Dashboard Overview</h1>
<div class="grid grid-cols-3 gap-4">
    <div class="p-4 bg-indigo-50 border border-indigo-200 rounded-md">
        <span class="text-xs text-indigo-700 font-semibold">Active Panels</span>
        <p class="text-2xl font-bold text-indigo-950 mt-1">14</p>
    </div>
</div>
{% endblock %}
```

The Result: `admin_dashboard.html` automatically gets the university header and footer from Level 1, the sidebar from Level 2, and renders its own stats into Level 3!

Pattern 2: Component Composition with `{% include %}`

If you have multiple independent reusable widgets or partials (e.g. a search bar, a panel summary card, an examiner table), you do not "extend" them—you **include** them:

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```
{% extends 'master_frame.html' %}

{% block body %}
<div class="max-w-6xl mx-auto py-8 px-4 space-y-6">
    <!-- Include Reusable Search Filter Bar -->
    {% include 'partials/_search_filters.html' %}

    <!-- Include Panel Details Card -->
    {% include 'partials/_panel_summary_card.html' %}

    <!-- Include Signatory Committee Table -->
    {% include 'partials/_committee_table.html' %}
</div>
{% endblock %}
```

Pattern 3: Jinja2 Macros (Reusable Functions with Arguments)

Macros are the Jinja2 equivalent of Python functions. They let you define reusable HTML components that accept dynamic parameters (labels, error messages, values):

```
<!-- components/form_macros.html -->
{% macro render_input(name, label, type='text', value='', error='') %}
<div class="mb-4">
    <label for="{{ name }}" class="block text-sm font-medium text-slate-700 mb-1">{{ label }}</label>
    <input type="{{ type }}" name="{{ name }}" id="{{ name }}" value="{{ value }}"
           class="w-full px-3 py-2 border rounded-md {% if error %}border-red-500 bg-red-50{% else %}bo
    {% if error %}
        <p class="text-xs text-red-600 mt-1">{{ error }}</p>
    {% endif %}
</div>
{% endmacro %}
```

In any child template, import and call the macro like a function:

```
{% extends 'master_frame.html' %}
{% from 'components/form_macros.html' import render_input %}

{% block body %}
<form method="POST" class="max-w-md mx-auto p-6">
    {{ render_input('username', 'Admin Username', value=username, error=errors.get('username')) }}
    {{ render_input('password', 'Password', type='password', error=errors.get('password')) }}
    <button type="submit" class="btn btn-primary">Sign In</button>
</form>
{% endblock %}
```

Pattern 4: Multiple Named Sockets in a Single Base Frame

Instead of having separate layout templates, define multiple specialized blocks inside `master_frame.html`:

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```
<!-- Inside master_frame.html: -->
<head>
    <title>SDSF Portal</title>
    {% block extra_head %}{# Page-specific CSS or CDN scripts #}{% endblock %}
</head>
<body>
    <header>...</header>
    <div class="flex">
        <aside>{% block sidebar %}{# Custom sidebar if provided #}{% endblock %}</aside>
        <main>{% block body %}{# Main view body #}{% endblock %}</main>
    </div>
    <footer>...</footer>
    {% block scripts %}{# Page-specific JS at bottom of body #}{% endblock %}
</body>
```

A child template can inject content into all 4 blocks independently in one single file!

Pattern 5: Conditional / Dynamic `{% extends %}`

Jinja2 allows the target of `{% extends %}` to be a dynamic Python variable or ternary expression:

```
{# Dynamically switch base frame if rendered inside an iframe/modal vs standalone page: #}
{% extends 'modal_frame.html' if is_modal else 'master_frame.html' %}
```

Architectural Decision Matrix: When to Use Which Pattern

Pattern	Syntax	Primary Purpose	Example in SDSF System
Single Inheritance	`{% extends 'master_frame.html' %}`	Standard stand-alone page inheriting global site layout.	`index.html`, `admin_login_page.html`
Chained Inheritance	`{% extends 'admin_base.html' %}`	Sub-system portal with persistent sidebar & sub-navigation.	`admin_dashboard.html`, `manage_teachers.html`
Component Include	`{% include 'partials/_card.html' %}`	Inserting reusable, static or context-dependent HTML chunks.	`footer.html`, `_flash_messages.html`
Jinja2 Macro	`{% macro %}` / `{% import %}`	Parameterized components that generate HTML from inputs.	Form inputs, modal dialogs, status badges.

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Module 3: Dynamic URL Resolution & Routing (`url_for`)

Hardcoding URLs such as `<a href="/admin/login">` is an anti-pattern. If you ever change the route path, every hardcoded link in your templates breaks. Flask provides `url_for()` to reverse-lookup URLs based on function names.

3.1 Syntax, Anatomy & Blueprint Resolution

```
# Normal route inside app.py:
url_for('home')                    # Returns: "/"

# Route inside a Blueprint:
url_for('admin.admin_login')       # Returns: "/admin/login"
#        ▲     ▲
#        |     └── View function name (def admin_login():)
#        └──────── Blueprint name (Blueprint('admin', ...))
```

3.2 Dynamic Parameters & Static Asset Linking

```
# Dynamic route in Python:
@panel_bp.route('/view/<int:panel_id>')
def view_panel(panel_id):
    ...

# Dynamic URL in Jinja2:
<a href="{{ url_for('panel.view_panel', panel_id=14) }}">View Panel 14</a>
# Output: /admin/view/14

# Static Assets (Images/CSS):
<img src="{{ url_for('static', filename='photo/davvLogo.png') }}" alt="DAVV Logo">
# Output: /static/photo/davvLogo.png
```

Module 4: Modular Architecture with Flask Blueprints

As applications grow, placing all routes in `app.py` creates unmaintainable code. Blueprints allow you to organize routes into decoupled modules.

4.1 Defining Blueprint Controllers

```
from flask import Blueprint, render_template, request, redirect, url_for

# 1. Instantiate the Blueprint
admin_bp = Blueprint('admin', __name__)

# 2. Attach routes to the blueprint instance
@admin_bp.route('/login', methods=['GET', 'POST'])
def admin_login():
    if request.method == 'POST':
        return redirect(url_for('admin.dashboard'))
    return render_template('admin_login_page.html')
```


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4.2 The Mathematical Derivation of `url_prefix`

```
from flask import Flask
from routes.admin import admin_bp

app = Flask(__name__)

# Register the Blueprint with a URL Prefix
app.register_blueprint(admin_bp, url_prefix='/admin')
```

The URL Prefix Math:

Route Defined in Blueprint	url_prefix in app.py	Final Browser URL
`/login`	`/admin`	`/admin/login`
`/dashboard`	`/admin`	`/admin/dashboard`
`/`	`/admin`	`/admin/`

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Module 5: HTTP Methods & Form Handling (GET vs. POST)

Web applications communicate using HTTP verbs. For database applications, the two most critical methods are `GET` and `POST`.

5.1 The Contract: GET vs. POST Data Transmission

Feature	GET	POST
Purpose	Retrieve data from server (safe/idempotent)	Submit data to mutate state (insert, update)
Data Location	URL query string (`?key=value`)	HTTP Request Body (payload)
Visibility	Visible in browser history & URL	Hidden from URL; encrypted in transit with SSL
Flask Access	`request.args.get('key')`	`request.form.get('key')`

5.2 The Post-Redirect-Get (PRG) Pattern

> **Architecture Rule:**
> Never return `render_template()` directly after a successful `POST` submission! If the user refreshes the page, the browser re-submits the form data, causing duplicate database insertions. Always use `return redirect(url_for(...))`. This forces the browser to issue a clean `GET` request to the target page.

Module 6: Session Management & User Authentication

HTTP is fundamentally **stateless**. This means the web server treats every incoming HTTP request as completely independent and has zero built-in memory of any previous requests. To build a secure, persistent administrative portal for SDSF Exam Cell, we use **Flask Sessions**.

6.1 Why Sessions Exist: The Stateless HTTP Dilemma

Consider the natural lifecycle of two consecutive clicks in a browser:

1. **Request 1 (Login):** The user visits `/admin/login`, submits their username and password via POST, and the database verifies they are valid. The server sends back a response and immediately terminates the TCP connection.
2. **Request 2 (Dashboard):** Two seconds later, the user clicks a link to visit `/admin/dashboard`. The browser opens a new TCP connection and sends a GET request.

> **The Stateless Dilemma:**
> On Request 2, the web server has complete amnesia. It has no native way of knowing that Request 2 came from the same human who just successfully authenticated during Request 1. Without a session mechanism, the user would be forced to re-enter their username and password on *every single link and button click*.

A **Session** solves this dilemma by establishing a persistent, remembered state between a specific browser and the server across multiple independent HTTP requests.

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6.2 How Flask Sessions Work Under the Hood (HMAC Signed Cookies vs. Server-Side Storage)

Syntactically, Flask represents the session as a familiar Python dictionary: `session['admin_id'] = admin['id']`. However, what Flask does under the hood differs from many other frameworks:

1. The Signed Client-Side Cookie Pattern (Flask Default)

1. **Serialization:** When you assign data to the `session` dictionary, Flask serializes the dictionary into a compact JSON string.
2. **Cryptographic Signing (HMAC-SHA256):** Flask takes that serialized string and hashes it against your application's `app.secret_key` using HMAC-SHA256, generating a tamper-proof cryptographic signature.
3. **Cookie Transmission:** Flask packages the data and signature into a browser cookie named `session` and sends it to the user's browser via the HTTP `Set-Cookie` header.
4. **Subsequent Requests:** On every future request, the browser automatically includes this `session` cookie. Flask splits the data from the signature and recalculates the HMAC hash using its `app.secret_key`. If the signatures match exactly, Flask loads the data into the `session` object for your view functions.

> **CRITICAL SECURITY DISTINCTION: "SIGNED" DOES NOT MEAN "ENCRYPTED"!**
>
> By default, Flask session cookies are **cryptographically signed, NOT encrypted**. This means anyone with browser DevTools can inspect and read the base64-decoded contents of the cookie. However, they **CANNOT modify or tamper with it** (e.g. changing `admin_id: 2` to `admin_id: 1` invalidates the HMAC signature, causing Flask to instantly reject the cookie).
>
> **The Golden Backend Rule:** NEVER store sensitive data like raw passwords, database connection strings, or credit cards in the Flask `session`! Store only non-sensitive identifiers such as `user_id`, `username`, or `role`.

2. The Crucial Role of `app.secret_key`

Without a secret key, cryptographic signing is impossible. If you attempt to use `session` or `flash()` without configuring a secret key, Flask immediately raises:

```
RuntimeError: The session is unavailable because no secret key was set.
Set the secret_key on the application to something unique and secret.
```

Always load the secret key from an environment variable (`.env`) with a development fallback:

```
import os
app.secret_key = os.getenv("SECRET_KEY", "sdsf-exam-cell-dev-secret-key-2026")
```

6.3 Complete Session Operations Reference (`set`, `get`, `pop`, `clear` & Expiration)

The Flask `session` object supports all standard Python dictionary operations. Below is the complete reference guide:

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Operation	Code Syntax	Behavior & Return Value	Best Use Case
Set Value	`session['admin_id'] = admin['id']`	Stores the key-value pair in the session dictionary for this client.	User authentication, saving active program or semester selection.
Safe Read (Recommended)	`admin_id = session.get('admin_id')`	Returns the stored value if present; returns `None` (or default) if absent. **Never raises KeyError.**	Checking authentication status in routes or templates.
Direct Read	`admin_id = session['admin_id']`	Returns the value if present; raises `KeyError` if absent.	Only when you are 100% certain the user is already authenticated.
Membership Check	`if 'admin_id' in session:`	Returns `True` if the key exists, `False` otherwise.	Fast conditional route guards and Jinja2 navbar logic.
Remove Key	`session.pop('temp_filter', None)`	Removes the specific key from the session and returns its value.	Clearing temporary wizard form steps or one-time tokens.
Full Logout (Clear All)	`session.clear()`	Purges all keys and values from the session dictionary completely.	User logout route (`/admin/logout`).

Session Lifespan & Expiration Modes

- **Browser-Session Cookie (Default):** If `session.permanent = False` (the default), the cookie is destroyed the moment the user completely closes their browser window.
- **Permanent Session Cookie:** If set to `session.permanent = True`, the cookie persists across browser restarts for a configurable duration (default: 31 days):

```
from datetime import timedelta
app.permanent_session_lifetime = timedelta(days=7)  # Keeps admin logged in for 7 days

@admin_bp.route('/login', methods=['POST'])
def admin_login():
    # ... after verifying admin ...
    session.permanent = True
    session['admin_id'] = admin['id']
    return redirect(url_for('admin.admin_dashboard'))
```

6.4 Flash Messaging Mechanics: Temporary Session Stashes

Beginners often treat `flash()` as a standalone feature, but it is actually a specialized wrapper around the Flask `session`:

1. When you call `flash("Panel approved!", "success")`, Flask creates an internal list at `session['_flashes']` and appends `("success", "Panel approved!")`.
2. Because it is written into the session cookie, the message safely survives a `redirect()`!
3. When the destination template calls `get_flashed_messages()`, Flask reads the messages and **immediately deletes `_flashes` from the session**. This guarantees that messages are displayed exactly once and never reappear on page refresh.

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6.5 Route Protection with Custom Decorators (`@login_required`) & Complete Lifecycle

In web application engineering, route security and session lifecycle management ensure that unauthorized users cannot view confidential administrative records. Manually writing `if 'admin_id' not in session: return redirect(...)` inside every single route violates the DRY (Don't Repeat Yourself) principle. The industry standard is to wrap protected views in a reusable Python decorator.

1. Demystifying Python Decorators & The Inner Mechanics of `@login_required`

In Python, functions are first-class objects—they can be passed as arguments, assigned to variables, and wrapped by other functions. When you attach `@login_required` above a route:

```
@admin_bp.route('/dashboard')
@login_required
def admin_dashboard():
    return render_template('admin_dashboard.html')
```

Python internally executes the following statement during application startup:

```
# The @ syntax is syntactic sugar for wrapper re-assignment:
admin_dashboard = login_required(admin_dashboard)
```

Here is the production-grade implementation with deep line-by-line commentary:

```
from functools import wraps
from flask import session, redirect, url_for, flash, request

def login_required(f):
    """
    Production-Grade Route Guard Decorator.
    Intercepts incoming HTTP requests before the view function executes.
    """
    @wraps(f)  # CRITICAL: Preserves the original function's name and identity!
    def decorated_function(*args, **kwargs):
        # 1. Inspect the cryptographically verified session dictionary
        if 'admin_id' not in session:
            # 2. Inform the user why they were kicked out
            flash("Unauthorized access. Please log in to view this confidential page.", "danger")

            # 3. Capture the URL they originally tried to visit (The 'next' parameter)
            return redirect(url_for('admin.admin_login', next=request.path))

        # 4. If admin_id IS present, allow execution to proceed into the view
        return f(*args, **kwargs)

    return decorated_function
```

> **THE CRITICAL `@wraps(f)` TRAP IN FLASK**
>
> Never omit `@wraps(f)` from `functools` when authoring Flask decorators!
>
> **Why it crashes without `@wraps(f)` :** Flask uses the decorated function's internal name (`__name__`) as its endpoint identifier for URL routing. Without `@wraps(f)` , Python renames *every single decorated function* to `"decorated_function"` . As soon as you decorate a second route, Flask crashes during startup with:

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```
AssertionError: View function mapping is overwriting an existing endpoint function: decorated_funct
```

`@wraps(f)` copies the original name (`'admin_dashboard'`), docstring, and module over to the wrapper, guaranteeing unique endpoint registration.

2. Preserving User Workflow: The "Next" URL Redirect Pattern

When an admin clicks a link to `/admin/panel/create` while logged out, dumping them onto the dashboard after login is frustrating. The industry standard is to preserve their intended target:

1. The `@login_required` decorator passes the current target path: `redirect(url_for('admin.admin_login', next=request.path))` .
2. The browser visits: `/admin/login?next=/admin/panel/create` .
3. Inside the POST handler of `admin_login()` , upon verifying credentials, read `request.args.get('next')` :

```
# Inside admin_login() after verifying admin:
session['admin_id'] = admin['id']
session['admin_user'] = admin['username']
flash("Welcome back, Administrator!", "success")

# Read intended destination:
next_page = request.args.get('next')

# SECURITY CHECK: Prevent Open Redirect Attacks!
# Ensure next_page is a relative path (starts with /), NOT an external malicious domain:
if not next_page or not next_page.startswith('/') or next_page.startswith('//'):
    next_page = url_for('admin.admin_dashboard')

return redirect(next_page)
```

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3. The Complete SDSF Exam Cell Authentication Lifecycle (Step-by-Step Machine)

Step / State	Actor	Under-the-Hood Actions	HTTP Transmission
1. Unauthenticated Visit	Guest Browser	Visits `/admin/dashboard` without credentials.	`GET /admin/dashboard` (No session cookie).
2. Route Interception	`@login_required`	Checks `'admin_id' in session` (Evaluates `False`). Generates flash alert.	`HTTP 302 Found` → `Location: /admin/login?next=/admin/dashboard`.
3. Credential Submission	Admin User	Fills login form. `validate_admin()` queries TiDB Cloud table.	`POST /admin/login` with form payload.
4. Session Genesis	Flask Controller	Assigns `session['admin_id'] = admin['id']`. Flask computes HMAC-SHA256 signature using `app.secret_key`.	`HTTP 302 Found` + `Set-Cookie: session=eyJhZG1p...`.
5. Authenticated Render	Jinja2 Engine	Browser automatically includes cookie. Global `session` is injected. Navbar reveals "Active: Admin" and "Logout".	`HTTP 200 OK` with confidential panel data rendered.
6. Termination (Logout)	Logout Controller	Executes `session.clear()`. Overwrites cookie with expired timestamp.	`HTTP 302 Found` + `Set-Cookie: session=; Expires=1970`.

4. Dynamic Header State Switching: "Admin Login" vs. "Active: Admin | Logout"

Flask automatically injects the active `session` object into Jinja2's global template environment for every request. You never need to pass authentication state as a route keyword argument!

In your persistent header (`master_frame.html` or `navbar.html`), evaluate `if 'admin_id' in session`:

```
<!-- Persistent University Header / Navbar -->
<div class="flex items-center gap-3">
    {% if 'admin_id' in session %}
        <!-- 1. LOGGED IN STATE: Pulsing status indicator + Admin name + Logout -->
        <div class="inline-flex items-center gap-1.5 px-2.5 py-1 rounded-full text-xs font-semibold bg-
            <span class="w-1.5 h-1.5 rounded-full bg-emerald-500 animate-pulse"></span>
            <span>Active: <strong>{{ session.get('admin_user', 'Admin') }}</strong></span>
        </div>

        <a href="{{ url_for('admin.admin_logout') }}"
           class="text-xs font-semibold text-rose-600 hover:text-rose-700 bg-rose-50 hover:bg-rose-100
            Logout
        </a>
    {% else %}
        <!-- 2. LOGGED OUT / GUEST STATE: Clean Login Button -->
        <a href="{{ url_for('admin.admin_login') }}"
           class="text-xs font-medium text-slate-700 hover:text-indigo-600 px-3 py-1.5 rounded-md borde
            Admin Login
        </a>
    {% endif %}
</div>
```


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5. The Industry Standard Production Logout Architecture (`/admin/logout`)

Where does the logout route live, and what must it execute under the hood? In professional engineering, logout is declared inside the controlling Blueprint (`routes/admin.py`):

```
# Inside routes/admin.py:
@admin_bp.route('/logout', methods=['GET'])
def admin_logout():
    """
    Standard Production Logout Controller:
    1. Wipes all server-side session variables.
    2. Instructs client browser to invalidate session cookie.
    3. Flashes a feedback notice.
    4. Redirects safely to the public login screen.
    """
    # 1. Clear all session keys (admin_id, admin_user, roles)
    session.clear()

    # 2. Flash a one-time exit message to confirm termination
    flash("You have been successfully logged out.", "info")

    # 3. Post-Redirect-Get pattern back to login
    return redirect(url_for('admin.admin_login'))
```

Detailed Comparison: `session.clear()` vs. `session.pop()` vs. `session['admin_id'] = None`

Method	Syntax	Security Assessment	Residual State Risk
`session.clear()`	`session.clear()`	**Gold Standard (Best Practice):** Purges the entire session dictionary and forces cookie recreation with zero keys.	**Zero Risk:** All temporary draft forms, wizard steps, and user tokens are completely obliterated.
`session.pop()`	`session.pop('admin_id', None)`	**Fragile:** Only deletes the specified key.	**High Risk:** Leaves behind `admin_user` , `panel_draft` , or privilege flags that other views might inadvertently trust.
`Key Assignment`	`session['admin_id'] = None`	**Dangerous Anti-Pattern:** The key `'admin_id'` still exists with a value of `None` !	**Critical Trap:** A check written as `if 'admin_id' in session:` still evaluates to `True` !

6. The "Browser Back Button" Security Vulnerability (BFCache) & Cache-Control Remedy

> **CRITICAL SECURITY RISK: The Post-Logout Back Button Trap**
> In university computer laboratories and shared faculty terminals, after an administrator clicks "Logout", the next person sitting at that workstation can press the browser's **Back Button** and view confidential examination panel sheets!
>
> **Why this happens:** The browser did not query the server; it retrieved an in-memory snapshot of the rendered DOM tree from its **Back-Forward Cache (BFCache)**.

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> **The Production Solution:** Force the browser to invalidate its local cache and re-query Flask on every navigation click by attaching HTTP Cache-Control headers in `app.py`:

```
# Inside app.py:
@app.after_request
def add_security_headers(response):
    """
    Prevents browsers from caching sensitive administrative screens.
    Forces browser to re-request from server on 'Back' button clicks.
    """
    response.headers['Cache-Control'] = 'no-cache, no-store, must-revalidate'
    response.headers['Pragma'] = 'no-cache'
    response.headers['Expires'] = '0'
    return response
```

How the Headers Protect the System:

- `no-store`: Commands the browser never to save any copy of the HTML page in memory, RAM, or disk storage.
- `no-cache` & `must-revalidate`: Commands the browser that it must send a new HTTP request to Flask before rendering anything.
- When the logged-out user presses "Back", the browser is forced to send a new `GET /admin/dashboard` request. The `@login_required` decorator intercepts the request, detects that `admin_id` is absent from `session`, and instantly kicks the browser back to the login page!

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6.6 The Complete Login & Logout Blueprint Architecture & Dynamic Navbar Switching (`master_frame.html`)

A frequent challenge for full-stack developers is coordinating the server-side authentication state with the visual layout—specifically, ensuring that navigation headers dynamically display **"Admin Login"** when logged out, and automatically switch to **"Active: Admin | Logout"** when authenticated.

1. The Architectural Choice: Manual `msg` Parameter vs. Global `session` Scope

When beginners try to pass login status to templates, they often attempt passing a message parameter (e.g. `render_template('page.html', msg="logged_in")`). Here is why that approach fails in production:

Architecture Pattern	How It Works	Major Weakness / Vulnerability	Production Verdict
Pattern A: Manual Variable Passing `render_template(..., msg="logged_in")`	Every single route must manually pass `msg="logged_in"` to its template view.	**High Fragility & DRY Violation:** If you create 15 views (e.g. `/panel/create`, `/teachers`, `/subjects`) and forget to pass `msg` in even one route, the navbar immediately loses state and displays "Admin Login" even though the user is authenticated.	**Anti-Pattern (Do Not Use)**
Pattern B: Global Session Context `{% if 'admin_id' in session %}`	Flask automatically injects the active `session` dictionary into every Jinja2 template context on every request.	**Zero Parameter Overhead:** You write zero Python parameters in view functions. The template inspects the cryptographically signed session cookie directly. Works automatically across 100% of routes!	**The Production Standard (Best Practice)**

2. Complete Standard Production Blueprint: `routes/admin.py`

Below is the complete, canonical implementation of the Admin Authentication Blueprint managing login, route protection, and logout:

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```
# routes/admin.py
from functools import wraps
from flask import Blueprint, render_template, request, redirect, flash, url_for, session
from databases.queries import validate_admin

admin_bp = Blueprint('admin', __name__)

# ============================================================
# 1. ROUTE GUARD DECORATOR
# ============================================================
def login_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if 'admin_id' not in session:
            flash("Unauthorized access. Please log in first.", "danger")
            return redirect(url_for('admin.admin_login', next=request.path))
        return f(*args, **kwargs)
    return decorated_function


# ============================================================
# 2. LOGIN ROUTE (GET & POST)
# ============================================================
@admin_bp.route('/login', methods=['GET', 'POST'])
def admin_login():
    # If already authenticated, redirect straight to dashboard:
    if 'admin_id' in session:
        return redirect(url_for('admin.admin_dashboard'))

    if request.method == 'POST':
        username = request.form.get('username', '').strip()
        password = request.form.get('password', '').strip()

        # Query TiDB database:
        admin = validate_admin(username, password)
        if admin:
            # 1. Store credentials in session:
            session['admin_id'] = admin['id']
            session['admin_user'] = admin['username']
            flash("Login successful! Welcome back.", "success")

            # 2. Resolve destination (Next URL or Dashboard):
            next_page = request.args.get('next')
            if not next_page or not next_page.startswith('/') or next_page.startswith('//'):
                next_page = url_for('admin.admin_dashboard')

            return redirect(next_page)
        else:
            return render_template("admin_login_page.html",
                                   error="Invalid username or password.",
                                   username=username)

    return render_template("admin_login_page.html")


# ============================================================
# 3. PROTECTED ADMIN DASHBOARD (GET)
```


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```
# ============================================================
@admin_bp.route('/dashboard', methods=['GET'])
@login_required  # Intercepts unauthenticated users automatically!
def admin_dashboard():
    # Notice: NO msg parameter is needed! master_frame.html reads session directly!
    return render_template("admin_dashboard.html")


# ============================================================
# 4. STANDARD LOGOUT CONTROLLER (GET)
# ============================================================
@admin_bp.route('/logout', methods=['GET'])
def admin_logout():
    # 1. Purge all session keys completely
    session.clear()

    # 2. Flash confirmation notification
    flash("You have been successfully logged out.", "info")

    # 3. Post-Redirect-Get pattern back to login screen
    return redirect(url_for('admin.admin_login'))
```

3. The Dynamic Master Frame Button Implementation (`master_frame.html`)

Inside `templates/master_frame.html`, the top-right header area checks `if 'admin_id' in session`. Because `master_frame.html` is the parent frame extended by all views, this button toggles everywhere across the entire application automatically:

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```
<!-- Inside templates/master_frame.html: Header Right Section -->
<div class="flex items-center gap-3 sm:gap-4">

    {% if 'admin_id' in session %}
        <!-- 1. SHOWN EXCLUSIVELY WHEN LOGGED IN -->
        <div class="hidden sm:inline-flex items-center gap-1.5 px-2.5 py-1 rounded-full text-xs font-se
            <span class="w-1.5 h-1.5 rounded-full bg-emerald-500 animate-pulse"></span>
            <span>Active: <strong>{{ session.get('admin_user', 'Admin') }}</strong></span>
        </div>

        <a
            href="{{ url_for('admin.admin_logout') }}"
            class="text-xs sm:text-sm font-semibold text-rose-600 hover:text-rose-700 bg-rose-50 hover:
            title="Terminate Active Admin Session"
        >
            Logout
        </a>
    {% else %}
        <!-- 2. SHOWN EXCLUSIVELY WHEN LOGGED OUT (PUBLIC GUEST) -->
        <a
            href="{{ url_for('admin.admin_login') }}"
            class="text-xs sm:text-sm font-medium text-slate-700 hover:text-indigo-600 px-3 py-1.5 roun
        >
            Admin Login
        </a>
    {% endif %}

    <!-- DAVV Institutional Logo -->
    <img
        src="{{ url_for('static', filename='photo/davvLogo.png') }}"
        alt="DAVV Logo"
        class="h-14 sm:h-16 w-auto object-contain"
    >
</div>
```

4. End-to-End Verification Flow

1. **Guest State:** User visits `/` (home page). `'admin_id' in session` is `False`. Top-right button displays **"Admin Login"**.
2. **Login Submission:** User visits `/admin/login` and enters valid credentials. The route runs `session['admin_id'] = admin['id']`.
3. **Authenticated State:** The user is redirected to `/admin/dashboard`. The top-right header automatically renders **"Active: admin"** and the red **"Logout"** button.
4. **Global Persistence:** The user clicks to any other page (e.g. `/`). The header *still* displays **"Active: admin | Logout"** because the signed session cookie persists across all routes.
5. **Logout Execution:** The user clicks **"Logout"**. The route runs `session.clear()` and redirects to `/admin/login`. The session cookie is purged, and the header flips back to displaying **"Admin Login"**.

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Module 7: Database Engineering: Architecture, Seeding & Complete CRUD

The core responsibility of the backend in the SDSF Exam Cell system is executing database operations against the TiDB Cloud relational database.

7.1 Database vs. Table: The Container Analogy

A frequent error encountered by developers is `(1049, "Unknown database 'name'")`. Understanding the relational hierarchy prevents this:

- **Database:** The top-level **container (folder)** on the server (e.g., `sdsf_result_db`). It must be explicitly created via `CREATE DATABASE name;` before any connection can attach to it.
- **Table:** The individual **data spreadsheets (files)** living inside that database (e.g., `admin`, `teachers`).
- **The `sys` Trap:** Connecting to `DATABASE=sys` triggers `(1142, "CREATE command denied")` because `sys` is MySQL/TiDB's protected internal metadata catalog. Always connect to a user database (such as `sdsf_result_db` or `test`).

7.2 Demystifying Connections, Cursors & `try...finally`

Database communication involves two distinct layers:

- **The Connection (`conn`):** The telephone wire between Python and TiDB Cloud. Opening it takes network time; keeping it open indefinitely causes server memory leaks.
- **The Cursor (`cursor`):** The courier/messenger that travels over that wire to execute SQL and bring back results.

Why `try...finally` is Critical: Preventing Ghost Connections

> **The Danger of Omission:**
> If you write `conn = get_connection()` and your query crashes due to bad SQL syntax, Python jumps out of the function immediately. Any `conn.close()` written below it is skipped! Over time, dozens of unclosed "ghost" connections stay alive on TiDB until the database server crashes from connection pool exhaustion.

```
conn = get_connection()
try:
    with conn.cursor() as cursor:
        cursor.execute("SELECT ...")
        return cursor.fetchone()
finally:
    # GUARANTEED EXECUTION:
    # No matter if the query succeeds, fails, crashes, or returns early,
    # the finally block ALWAYS executes to close the network connection.
    conn.close()
```

What Does `with conn.cursor() as cursor` Do?

`with` is a Python **Context Manager**. It replaces manual cleanup:

1. When entering the `with` block, it creates the cursor object.
2. When exiting the block (even upon crash or return), Python automatically invokes `cursor.close()`.

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7.3 Clean Query Architecture: Separate SQL Variables, Inline Queries & `--sql` / `f"""--sql"""` Syntax

In Python backend development, writing SQL queries cleanly, securely, and readably is a fundamental skill. Below are the standard patterns, editor tooling enhancements, and crucial security rules:

Pattern A: The Separate SQL Variable Pattern (Highly Recommended)

```
with conn.cursor() as cursor:
    # 1. Define SQL string in its own dedicated variable
    sql = "SELECT id, username FROM admin WHERE username = %s AND password = %s;"

    # 2. Execute passing the variable and tuple parameters
    cursor.execute(sql, (username, password))

    return cursor.fetchone()
```

Why Pattern A is Superior for Backend Developers:

1. **Effortless Debugging & Logging:** You can add `print(sql)` or `logger.debug(sql)` right above the execution line without altering the execution call.
2. **Clean, Readable Indentation:** The query string is not awkwardly nested inside parentheses and commas.
3. **Code Reuse:** The same `sql` string can be passed to multiple executions (e.g. batch or prepared statements).

Pattern B: The Inline Triple-Quote Style

```
with conn.cursor() as cursor:
    cursor.execute("""
        SELECT id, username
        FROM admin
        WHERE username = %s AND password = %s;
    """, (username, password))

    return cursor.fetchone()
```

When to use Pattern B: Useful for quick scripts or massive multi-join queries spanning 15+ lines where creating an intermediate variable is unnecessary.

Pattern C: The IDE Syntax Highlighting Tags & The MySQL `--` Whitespace Trap

Many professional developers format queries using an initial comment tag inside Python multi-line triple quotes so their code editor renders full SQL color syntax highlighting:

```
with conn.cursor() as cursor:
    # Notice the mandatory space after the two dashes:
    sql = """-- sql
        SELECT id, username, full_name
        FROM teachers
        WHERE employee_code = %s AND is_active = %s;
    """
    cursor.execute(sql, (emp_code, True))
    return cursor.fetchone()
```

Why Developers Write `-- sql`:

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- **Editor Tooling & Highlighting:** Modern IDEs (such as VS Code with plugins like *sql-template-literal* or *Inline SQL*, and JetBrains PyCharm) recognize the leading SQL comment marker. The editor dynamically switches its syntax engine from Python plain-text string mode to **native SQL highlighting**.
- **Visual Clarity:** Keywords like `SELECT`, `FROM`, `WHERE`, and `JOIN` light up in distinct SQL syntax colors directly inside your Python file, making query typos immediately visible before execution.

> **CRITICAL ENGINE TRAP: The MySQL / TiDB Double-Dash Whitespace Requirement**
> In MySQL and TiDB (Section 9.6 of the Official MySQL Manual), a single-line comment using `--` **MUST be followed by at least one whitespace character (space or tab)**.
>
> **What happens if you omit the space?**

```
# FATAL CRASH IN TiDB / MySQL:
sql = """--sql
    SELECT * FROM admin WHERE username=%s;
"""
# Raises: pymysql.err.ProgrammingError: (1064, "You have an error in your SQL syntax;
# check the manual that corresponds to your TiDB version for the right syntax near '--sql'")
```

> **Why this error happens:** Without a space after `--`, MySQL/TiDB does **NOT** parse it as a comment! Instead, it interprets the double-dash as unary minus operators or an illegal token, causing the parser to fail at character 1.

The 3 Production Solutions to Prevent Error 1064:

Pattern	Syntax Example	Status & Behavior
1. Single-Line Comment (With Space)	`sql = """--sql SELECT ..."""`	**Valid & Safe:** The space after `--` satisfies MySQL/TiDB comment syntax while triggering IDE highlighting.
2. Standard C-Style Block Comment	`sql = """/* sql */ SELECT ..."""`	**Universal Standard:** Works across 100% of SQL dialects (MySQL, TiDB, PostgreSQL, SQLite, Oracle) with zero whitespace sensitivity.
3. Clean Triple Quotes (No Tag)	`sql = """ SELECT ..."""`	**Pure Standard:** Eliminates all comments. Simple, clean, and universally supported.

Pattern D: The `f"""--sql ... """` Style: Severe Security Warning & The SQL Injection Trap

> **CRITICAL SECURITY WARNING: NEVER USE PYTHON F-STRINGS FOR USER INPUT VALUES IN SQL!**
> Beginners frequently combine Python f-strings with SQL (e.g., `f"""--sql SELECT * WHERE username = '{username}'"""`). This bypasses database parameterization and introduces devastating **SQL Injection (SQLi)** vulnerabilities.

Observe the difference between the vulnerable f-string approach and the safe parameterized standard:

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```
# ============================================================
# 1. DANGEROUS PATTERN (DO NOT USE FOR USER INPUT):
# ============================================================
# Python f-string directly concatenates raw strings before sending to TiDB:
sql = f"""--sql
    SELECT id, username FROM admin
    WHERE username = '{username}' AND password = '{password}';
"""
cursor.execute(sql)

# ATTACK SCENARIO:
# If an attacker enters the username:  admin' --
# The interpolated SQL executed on the server becomes:
# SELECT id, username FROM admin WHERE username = 'admin' -- ' AND password = '...';
# The '--' comments out the entire password check! The attacker is logged in instantly.


# ============================================================
# 2. SECURE PATTERN (THE PRODUCTION STANDARD):
# ============================================================
# Use triple quotes with '-- sql' (or '/* sql */') for IDE highlighting,
# but pass all user variables strictly via %s parameter placeholders:
sql = """-- sql
    SELECT id, username FROM admin
    WHERE username = %s AND password = %s;
"""
cursor.execute(sql, (username, password))
# Here, TiDB treats the user's input strictly as literal string values,
# preventing any injected quotes or comments from being executed as SQL commands.
```

When is `f"""--sql ... """` legitimately permissible?

Database parameterization with `%s` only works for *values* (data literals), NOT for SQL identifiers such as table names or column names (e.g., `SELECT * FROM %s` throws a syntax error in MySQL/TiDB). In rare advanced cases where dynamic table selection is required, an f-string may be used **only if strictly validated against a hardcoded whitelist**:

```
# SAFE USE OF f-string: Rigorous Whitelist Validation
ALLOWED_TABLES = {"programs", "teachers", "subjects"}

if table_choice not in ALLOWED_TABLES:
    raise ValueError(f"Security Exception: Table '{table_choice}' is not authorized.")

# Dynamic table identifier safely interpolated, while query values remain parameterized:
sql = f"""--sql
    SELECT * FROM `{table_choice}`
    WHERE is_active = %s;
"""
cursor.execute(sql, (1,))
```


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7.4 Complete Cursor Methods Reference (Return Types & Exact Behavior)

Below is the complete reference guide for all major cursor methods and properties in Python DB-API (PyMySQL / MySQL / TiDB), detailing exactly what each returns when records exist versus when they do not:

Method / Property	Description	What It Returns (When Found)	What It Returns (When Empty / Failed)
`cursor.fetchone()`	Fetches the very next single row from the query result set.	A single **Dictionary** (with `DictCursor`): `{'id': 1, 'username': 'admin'}`	Python's `None`
`cursor.fetchall()`	Fetches *all* remaining rows from the query result set in memory.	A **List of Dictionaries**: `[{'id': 1, 'name': 'AI'}, {'id': 2, 'name': 'DS'}]`	An **Empty List**: `[]`
`cursor.fetchmany(size=N)`	Fetches the next chunk of up to `N` rows (useful for pagination).	A **List of Dictionaries** up to length `N`: `[{...}, {...}]`	An **Empty List**: `[]` (if no rows left)
`cursor.lastrowid`	Property returning the auto-generated primary key ID of an `INSERT`.	Integer ID of the inserted row: e.g. `42`	`0` or `None` (if table has no AUTO_INCREMENT)
`cursor.rowcount`	Property returning the count of rows affected by INSERT/UPDATE/DELETE.	Integer count of modified rows: e.g. `3` (for 3 rows updated)	`0` (if no rows matched the WHERE condition)
`cursor.executemany(sql, params)`	Executes one SQL query repeatedly across a list of tuples (bulk operations).	Total integer count of affected rows across all items.	`0` (if param list was empty)
`cursor.scroll(value, mode)`	Moves the internal cursor pointer forward/backward over results.	`None` (pointer moves internally).	Raises `IndexError` if scrolled out of bounds.

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Code Examples for Every Cursor Method:

```
# 1. fetchone() - Checking single login row:
cursor.execute("SELECT id, username FROM admin WHERE id = %s;", (1,))
admin = cursor.fetchone()
if admin:
    print("Found admin:", admin['username'])
else:
    print("Admin does not exist (returned None)")

# 2. fetchall() - Fetching all semester subjects:
cursor.execute("SELECT subject_code, subject_name FROM subjects WHERE semester = %s;", (5,))
subjects = cursor.fetchall()
# subjects is a list: [{'subject_code': 'DS6A-301', 'subject_name': 'Data Mining'}, ...]
for sub in subjects:
    print(sub['subject_code'], sub['subject_name'])

# 3. fetchmany(size) - Paginating large result sets:
cursor.execute("SELECT * FROM student_records;")
first_page = cursor.fetchmany(10)  # Fetches exactly 10 rows
second_page = cursor.fetchmany(10) # Fetches next 10 rows

# 4. lastrowid - Getting the ID of a newly inserted Exam Panel:
cursor.execute("INSERT INTO exam_panels (semester, academic_session) VALUES (%s, %s);", (5, '2025-26'))
new_panel_id = cursor.lastrowid
print("Created Panel with ID:", new_panel_id) # e.g., 14

# 5. rowcount - Verifying if an update succeeded:
cursor.execute("UPDATE exam_panels SET status = 'Approved' WHERE id = %s;", (14,))
if cursor.rowcount > 0:
    print("Status updated successfully!")
else:
    print("No panel found with ID 14 (0 rows affected)")

# 6. executemany() - Bulk inserting 50 subject rows in 1 single network call:
subject_rows = [
    (new_panel_id, 1, 101, 'RE', 5, 5),
    (new_panel_id, 2, 102, 'RE', 8, 8),
    (new_panel_id, 3, 103, 'RE', 9, 9)
]
insert_sql = "INSERT INTO exam_panel_subjects (panel_id, serial_no, subject_id, student_category, exami
cursor.executemany(insert_sql, subject_rows)
print(f"Bulk inserted {cursor.rowcount} subject rows in one call!")
```

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7.5 Database Seeding Without Credential Leaks

> **Security Alert (CWE-798):**
> Never write hardcoded passwords in `schema.sql` (e.g., `INSERT INTO admin VALUES ('admin', 'pass123');`). Because `schema.sql` is tracked in Git, your credentials will leak to GitHub!

The Secure Seeding Workflow:

1. Store admin credentials in your local `.env` file (which is git-ignored):

```
INITIAL_ADMIN_USER=admin
INITIAL_ADMIN_PASS=AdminSecret@2026
```

2. In `schema.sql`, only maintain the table structure:

```
CREATE TABLE IF NOT EXISTS admin (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    password VARCHAR(200) NOT NULL
);
```

3. In `init_db.py`, dynamically read `os.getenv("INITIAL_ADMIN_USER")` and insert the user only if it does not already exist.

7.6 Complete Production CRUD Operations

A. Create (INSERT with Parameterized Queries)

```
def add_teacher(employee_code, full_name, contact_no):
    with get_connection() as conn:
        with conn.cursor() as cursor:
            # ALWAYS use %s placeholders to prevent SQL Injection
            sql = """
                INSERT INTO teachers (employee_code, full_name, contact_no)
                VALUES (%s, %s, %s);
            """
            cursor.execute(sql, (employee_code, full_name, contact_no))
            conn.commit()
            return cursor.lastrowid
```

B. Read (SELECT Operations)

```
def get_subjects_by_semester(program_id, semester):
    with get_connection() as conn:
        with conn.cursor() as cursor:
            sql = """
                SELECT id, subject_code, subject_name
                FROM subjects
                WHERE program_id = %s AND semester = %s
                ORDER BY subject_code ASC;
            """
            cursor.execute(sql, (program_id, semester))
            return cursor.fetchall()  # Returns list of dicts: [{'subject_code': ...}, ...]
```


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C. Update (UPDATE Operation)

```
def update_panel_status(panel_id, new_status):
    with get_connection() as conn:
        with conn.cursor() as cursor:
            sql = "UPDATE exam_panels SET status = %s WHERE id = %s;"
            cursor.execute(sql, (new_status, panel_id))
            conn.commit()
            return cursor.rowcount > 0
```

D. Delete (DELETE Operation with Cascading Integrity)

```
def delete_panel(panel_id):
    with get_connection() as conn:
        with conn.cursor() as cursor:
            # Foreign key ON DELETE CASCADE automatically purges panel subjects & committee rows!
            sql = "DELETE FROM exam_panels WHERE id = %s;"
            cursor.execute(sql, (panel_id,))
            conn.commit()
```

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Module 8: Backend Error Handling & Defensive Programming

Robust backend systems handle unexpected inputs gracefully rather than crashing with 500 Internal Server Errors.

8.1 Custom Application Error Handlers

```
@app.errorhandler(404)
def page_not_found(e):
    return render_template('errors/404.html'), 404

@app.errorhandler(500)
def internal_server_error(e):
    # Log stack trace for administrators
    app.logger.error(f"Server Error: {e}")
    return render_template('errors/500.html'), 500
```

8.2 Form Validation & Error Propagation: `render_template` vs. `redirect` (Flash)

When handling user input errors (e.g. invalid credentials, duplicate usernames, passwords too short), choosing between `render_template()` and `redirect()` is one of the most critical architectural decisions:

Dimension	Pattern A: In-Place Error (`render_template`)	Pattern B: Transient Flash Alert (`redirect` + `flash`)
HTTP Status Code	`200 OK` (or `400 Bad Request` / `422 Unprocessable Entity`)	`302 Found` (Redirect to destination route)
Lifecycle	Synchronous (Same HTTP request cycle).	Asynchronous / Cross-Request (Saved to session cookie, consumed on next request).
Preserves Form Data?	**YES:** You pass `form_data=request.form` back to the template so the user does NOT lose what they typed.	**NO:** A redirect causes the browser to issue a fresh `GET` request, wiping all typed form inputs completely!
Best Use Case	**Form Validation Failures:** Username already exists, passwords do not match, missing fields.	**Post-Action Alerts:** Unauthorized access warnings (`@login_required`), session expired, successful mutations.

Deep Dive: In-Place Form Errors with `render_template()`

When user input fails validation, **never redirect**. Re-render the template immediately, passing both the error message and the submitted form data:

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```
# Inside routes/admin.py:
@admin_bp.route("/registration", methods=['GET', 'POST'])
def admin_register():
    if request.method == 'POST':
        username = request.form.get('username', '').strip()
        password = request.form.get('password', '').strip()

        # 1. Validation: Field presence
        if not username or not password:
            return render_template("admin_registration_form.html",
                                   error="All fields are required.",
                                   username=username)

        # 2. Validation: Business logic (Unique username check)
        if check_username_exists(username):
            return render_template("admin_registration_form.html",
                                   error="This username is already taken. Please choose another.",
                                   username=username)

        # 3. Success: Insert and Redirect (PRG Pattern)
        insert_in_admin(username, password)
        flash("Admin registered successfully! Please log in.", "success")
        return redirect(url_for('admin.admin_login'))

    return render_template("admin_registration_form.html")
```

Dictionary of Field-Level Errors (Multi-Field Validation)

If an application validates multiple form fields simultaneously (e.g. username taken AND password under 8 characters), pass a dictionary of errors:

```
errors = {}
if check_username_exists(username):
    errors['username'] = "Username already exists."
if len(password) < 8:
    errors['password'] = "Password must be at least 8 characters long."

if errors:
    # Pass entire dictionary to template:
    return render_template("admin_registration_form.html",
                           errors=errors,
                           form_data=request.form)
```

In HTML: Display inline red borders and field-level error text:

```
<div>
    <label for="username">Username</label>
    <input type="text" name="username" value="{{ form_data.get('username', '') }}"
           class="w-full px-3 py-2 border rounded-md {% if errors.get('username') %}border-red-500 bg-r
    {% if errors.get('username') %}
        <p class="text-xs text-red-600 mt-1">{{ errors['username'] }}</p>
    {% endif %}
</div>
```


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8.3 Unified Template Error Handling: Rendering In-Place Errors & Flash Messages Simultaneously

In real-world applications, a template often needs to display **both types of notifications simultaneously**:

- **Flash Messages:** Arriving from previous redirects (e.g. "*Unauthorized access. Please log in first*" from `@login_required`).
- **In-Place Errors:** Passed directly from the current view (e.g. `error="Invalid username or password"`).

The Unified Universal Alert Block Pattern

Place this consolidated block right at the top of your forms or inside `master_frame.html`:

```
<!-- ========================================================== -->
<!-- UNIFIED ALERT NOTIFICATION COMPONENT                       -->
<!-- Handles both Flash Messages AND In-Place Error Variables  -->
<!-- ========================================================== -->

{# 1. Check for Transient Flash Messages (arriving across redirects) #}
{% with messages = get_flashed_messages(with_categories=true) %}
    {% if messages %}
        <div class="space-y-2 mb-4">
            {% for category, message in messages %}
                <div class="p-3 rounded-lg text-sm font-medium border flex items-center gap-2
                    {% if category == 'danger' or category == 'error' %}bg-rose-50 text-rose-800 border
                    {% elif category == 'success' %}bg-emerald-50 text-emerald-800 border-emerald-200
                    {% elif category == 'warning' %}bg-amber-50 text-amber-800 border-amber-200
                    {% else %}bg-indigo-50 text-indigo-800 border-indigo-200{% endif %}">
                    <span>{{ message }}</span>
                </div>
            {% endfor %}
        </div>
    {% endif %}
{% endwith %}

{# 2. Check for Single In-Place Error (passed via render_template) #}
{% if error %}
    <div class="mb-4 p-3 rounded-lg bg-rose-50 border border-rose-200 text-rose-700 text-sm flex items-
        <svg class="w-4 h-4 text-rose-500 shrink-0" fill="none" stroke="currentColor" viewBox="0 0 24 2
            <path stroke-linecap="round" stroke-linejoin="round" stroke-width="2" d="M12 8v4m0 4h.01M21
        </svg>
        <span>{{ error }}</span>
    </div>
{% endif %}

{# 3. Check for Legacy or General Message (passed via render_template(..., msg=...)) #}
{% if msg and msg != error %}
    <div class="mb-4 p-3 rounded-lg bg-blue-50 border border-blue-200 text-blue-700 text-sm">
        {{ msg }}
    </div>
{% endif %}
```

Why this unified pattern is unbeatable:

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1. If an admin tries to open `/admin/dashboard`, they get redirected to `/admin/login` with a **flash danger** message ("*Unauthorized access*"). Section #1 renders it cleanly.
2. If they enter the wrong password, the route re-renders `admin_login_page.html` with `error="Invalid credentials"`. Section #2 catches and displays it in-place.
3. If a route passes a status string via `msg="..."`, Section #3 handles it gracefully without breaking.

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8.4 The 5 Commandments of Backend Engineering for This Project

1. **Never Trust Client Input:** Validate every `request.form` field on the server side, even if HTML5 `required` attributes exist.
2. **Always Use Parameterized SQL:** Never format strings into SQL queries (`f"SELECT * WHERE name = '{name}'"`). Always pass parameters via tuples to prevent SQL injection.
3. **Use Atomic Database Transactions:** When inserting a panel and its subject rows, wrap them in a single transaction. Call `conn.commit()` only when every row succeeds, and `conn.rollback()` if any row fails.
4. **Separate DDL from DML:** Keep table creation statements in `schema.sql` and operational queries in `queries.py` . Never run `CREATE TABLE` inside normal web requests.
5. **Protect Confidential Documents:** Implement role-based access checks so that only authenticated faculty and examination coordinators can view or print confidential examiner sheets.

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M.Tech Artificial Intelligence and Data Science • School of Data Science and Forecasting, Devi Ahilya Vishwavidyalaya, Indore
Portfolio: joyobrata-das.vercel.app • GitHub: github.com/holycompile • LinkedIn: Joyobrata Das • Contact: +91 8910366038
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